Worksheet 6.3 — Integration of Trig and Exponential Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 6: Integral Calculus

The Big Ideas

Reverse the trig, exponential, and log derivatives, including the linear-substitution pattern.

- $\int \sin x\,dx = -\cos x + C$, $\int \cos x\,dx = \sin x + C$, $\int e^x\,dx = e^x + C$, $\int \frac{1}{x}\,dx = \ln|x| + C$.

Worked Examples

Example 1 — Basic combination

Find $\int (5\cos x + 3e^x)\,dx$.

Solution. $5\sin x + 3e^x + C$.

Conclusion: $5\sin x + 3e^x + C$.

Example 2 — Exponential substitution

Find $\int e^{2x+7}\,dx$.

Solution. Divide by 2: $\dfrac{e^{2x+7}}{2}$.

Conclusion: $\dfrac{e^{2x+7}}{2} + C$.

Example 3 — Trig substitution

Find $\int \cos(3x-1)\,dx$.

Solution. Divide by 3: $\dfrac{\sin(3x-1)}{3}$.

Conclusion: $\dfrac{\sin(3x-1)}{3} + C$.

Example 4 — Definite integral, combined types

Evaluate $\int_0^{\pi/2} (3\sin x + \cos x)\,dx$ exactly.

Solution. Antiderivative $-3\cos x + \sin x$; at $\pi/2$: $0 + 1 = 1$; at 0: $-3 + 0 = -3$.

Conclusion: $1 - (-3) = 4$.

Example 5 — Logarithmic-form integral with linear substitution

Find $\displaystyle\int \frac{3}{2x-5}\,dx$.

Solution. Divide by $a=2$: $\dfrac{3\ln|2x-5|}{2}$.

Conclusion: $\dfrac{3}{2}\ln|2x-5|+C$.

Example 6 — Sine substitution with a negative coefficient

Find $\displaystyle\int \sin(4-3x)\,dx$ (note the negative inner coefficient).

Solution. Inner derivative is -3: divide by -3: $-\dfrac{\cos(4-3x)}{-3}$.

Conclusion: $\dfrac{\cos(4-3x)}{3}+C$.

Example 7 — Definite integral requiring exact special angles

Evaluate $\displaystyle\int_0^{\pi/3} \sin x\,dx$ exactly.

Solution. Antiderivative $-\cos x$; at $\pi/3$: $-\frac{1}{2}$; at 0: -1.

Conclusion: $-\frac{1}{2}-(-1)=\frac{1}{2}$.

Example 8 — Combining exponential and logarithmic integration

Find $\displaystyle\int \left(e^{4x}+\frac{2}{x}\right)dx$.

Solution. $\dfrac{e^{4x}}{4}+2\ln|x|$.

Conclusion: $\dfrac{e^{4x}}{4}+2\ln|x|+C$.

Example 9 — Application — accumulated oscillating rate

A tide's rate of change is $r(t)=2\cos\left(\dfrac{\pi}{6}t\right)$ m/hr. Find the net change in depth from $t=0$ to $t=3$.

Solution. $\int_0^3 2\cos(\pi t/6)\,dt=\left[\dfrac{12}{\pi}\sin(\pi t/6)\right]_0^3=\dfrac{12}{\pi}\sin(\pi/2)-0$.

Conclusion: net change $=\dfrac{12}{\pi}\approx 3.82$ m.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $\int (4\sin x - e^x)\,dx$.

Question 2

Find $\int e^{5x-3}\,dx$.

Question 3

Find $\int \sin(2x + \pi/3)\,dx$.

Question 4

Find $\int \dfrac{1}{5x+2}\,dx$.

Question 5 — Challenge

Evaluate $\displaystyle\int_0^{\pi} (2\cos x - \sin x)\,dx$ exactly.

Question 6

Find $\int (e^{3x} - \cos(2x) + 5)\, dx$.

Question 7

Find $\int \dfrac{4}{3x - 7}\, dx$.

Question 8

Find $\int \cos(5 - 2x)\, dx$ (note the negative inner coefficient).

Question 9

Evaluate $\displaystyle\int_0^{\pi/4} \cos x\, dx$ exactly.

Question 10

Find $\int \left(e^{2x} + \frac{3}{x}\right) dx$.

Question 11 — Challenge

A tide's rate of change is $r(t) = 3\sin\left(\frac{\pi}{6}t\right)$ m/hr; find the net change in depth from $t = 0$ to $t = 6$.

Question 12 — Challenge

Evaluate $\int_0^{\pi/6} 2\sin x \, dx$ exactly.

Question 13 — Challenge

Find $\int \left(\frac{5}{2x+1} - e^{-x}\right) dx$ (careful with the sign on the exponential term).

Question Recap & Answer Key

Question Recap

1. Find $\int (4\sin x - e^x)\,dx$.
2. Find $\int e^{5x-3}\,dx$.
3. Find $\int \sin(2x+\pi/3)\,dx$.
4. Find $\int \dfrac{1}{5x+2}\,dx$.
5. Evaluate $\displaystyle\int_0^{\pi} (2\cos x - \sin x)\,dx$ exactly.
6. Find $\int (e^{3x} - \cos(2x) + 5)\,dx$.
7. Find $\int \dfrac{4}{3x-7}\,dx$.
8. Find $\int \cos(5-2x)\,dx$ (note the negative inner coefficient).
9. Evaluate $\displaystyle\int_0^{\pi/4} \cos x\,dx$ exactly.
10. Find $\int \left(e^{2x} + \dfrac{3}{x}\right) dx$.
11. A tide's rate of change is $r(t) = 3\sin\left(\dfrac{\pi}{6}t\right)$ m/hr; find the net change in depth from $t=0$ to $t=6$.
12. Evaluate $\displaystyle\int_0^{\pi/6} 2\sin x\,dx$ exactly.
13. Find $\int \left(\dfrac{5}{2x+1} - e^{-x}\right) dx$ (careful with the sign on the exponential term).

✓ Answer Key

1. $-4\cos x - e^x + C$
2. $\dfrac{e^{5x-3}}{5} + C$
3. $-\dfrac{\cos(2x+\pi/3)}{2} + C$
4. $\dfrac{\ln|5x+2|}{5} + C$
5. antideriv $2\sin x + \cos x$; at π: $0 - 1 = -1$; at 0: $0 + 1 = 1$; result $= -1 - 1 = -2$
6. $\dfrac{e^{3x}}{3} - \dfrac{\sin(2x)}{2} + 5x + C$
7. $\dfrac{4\ln|3x-7|}{3} + C$
8. inner derivative -2: $-\dfrac{\sin(5-2x)}{-2} = \dfrac{\sin(5-2x)}{2} + C$
9. $[\sin x]_0^{\pi/4} = \frac{\sqrt{2}}{2} - 0 = \frac{\sqrt{2}}{2}$
10. $\dfrac{e^{2x}}{2} + 3\ln|x| + C$
11. $\left[-\dfrac{18}{\pi}\cos(\pi t/6)\right]_0^6 = -\dfrac{18}{\pi}\cos(\pi) - \left(-\dfrac{18}{\pi}\cos 0\right) = \dfrac{18}{\pi} + \dfrac{18}{\pi} = \dfrac{36}{\pi} \approx 11.46$ m
12. $[-2\cos x]_0^{\pi/6} = -2(\frac{\sqrt{3}}{2}) - (-2) = 2 - \sqrt{3}$
13. $\dfrac{5}{2}\ln|2x+1| + e^{-x} + C$ (the exponential term integrates to $+e^{-x}$ since $\int e^{-x}dx = -e^{-x}$, and we're subtracting that)